Current: charge flow rate

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “charge” to explain your model or result. Sentence starters are optional.

current: rate of charge flow

charge: a property responsible for electric interactions

interval: the time or range between two stated values

01 Bring back an earlier idea

Keep charge separation fixed and multiply one charge magnitude by 3.
Compare electric-force magnitude.

Word help: separation = distance between the specified objects or centers

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $I = \Delta q/\Delta t$.
Current is charge per time; 1 A = 1 C/s. Conventional current follows positive charge motion.

Worked reasoning: If 12 C pass a point in 4 s, average current is I=Q/t=3 A. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and label equal current before and after one resistor in steady state.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Draw conventional current around the closed circuit, outside the battery from + toward -. What happens if a wire is disconnected?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

12 C passes a point in 7 s.
Find I=12÷4.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Use the idea

At a junction, 5 A enters and 2 A leaves along one branch.
Find the other outgoing current in steady operation.
Explain the conservation rule.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

06 Choose a method

A siren emits 600 Hz steadily while approaching a stationary listener.
Compare observed and emitted frequency and explain.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A current of 2 A flows for 3 s. How much charge passes?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Construct two charge-time pairs that both give 2 A.
Explain why different total charges can give the same current.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3